Extension: Lengths in chords, secants, and tangents

Core Concepts: Extension: Lengths in chords, secants, and tangents

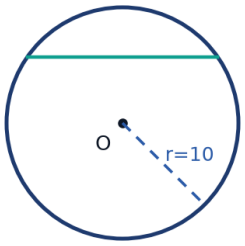
1 For the model $f(x) = 2x + 5$ in extension: lengths in chords, secants, and tangents, evaluate $f(2)$.

a. 9

b. 11

c. 4

d. 13



2 Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

3 A data set recorded for extension: lengths in chords, secants, and tangents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 5$ in extension: lengths in chords, secants, and tangents, evaluate $f(3)$.

a. 11

b. 13

c. 6

d. 15

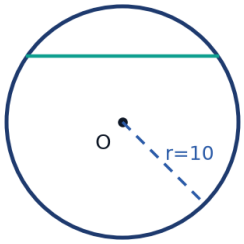
5 Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$



6 A data set recorded for extension: lengths in chords, secants, and tangents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 2x + 5$ in extension: lengths in chords, secants, and tangents, evaluate $f(4)$.

a. 13

b. 15

c. 8

d. 17

8 Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

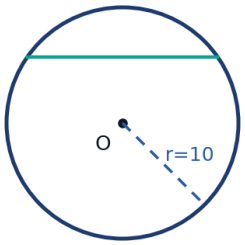
9 A data set recorded for extension: lengths in chords, secants, and tangents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5



Exam-Style Problem Solving & Modeling

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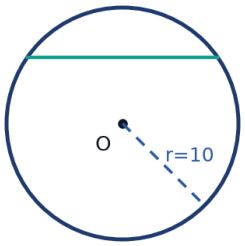
10 For the model $f(x) = 2x + 5$ in extension: lengths in chords, secants, and tangents, evaluate $f(5)$.

a. 15

b. 17

c. 10

d. 19



11 Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

12 A data set recorded for extension: lengths in chords, secants, and tangents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

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